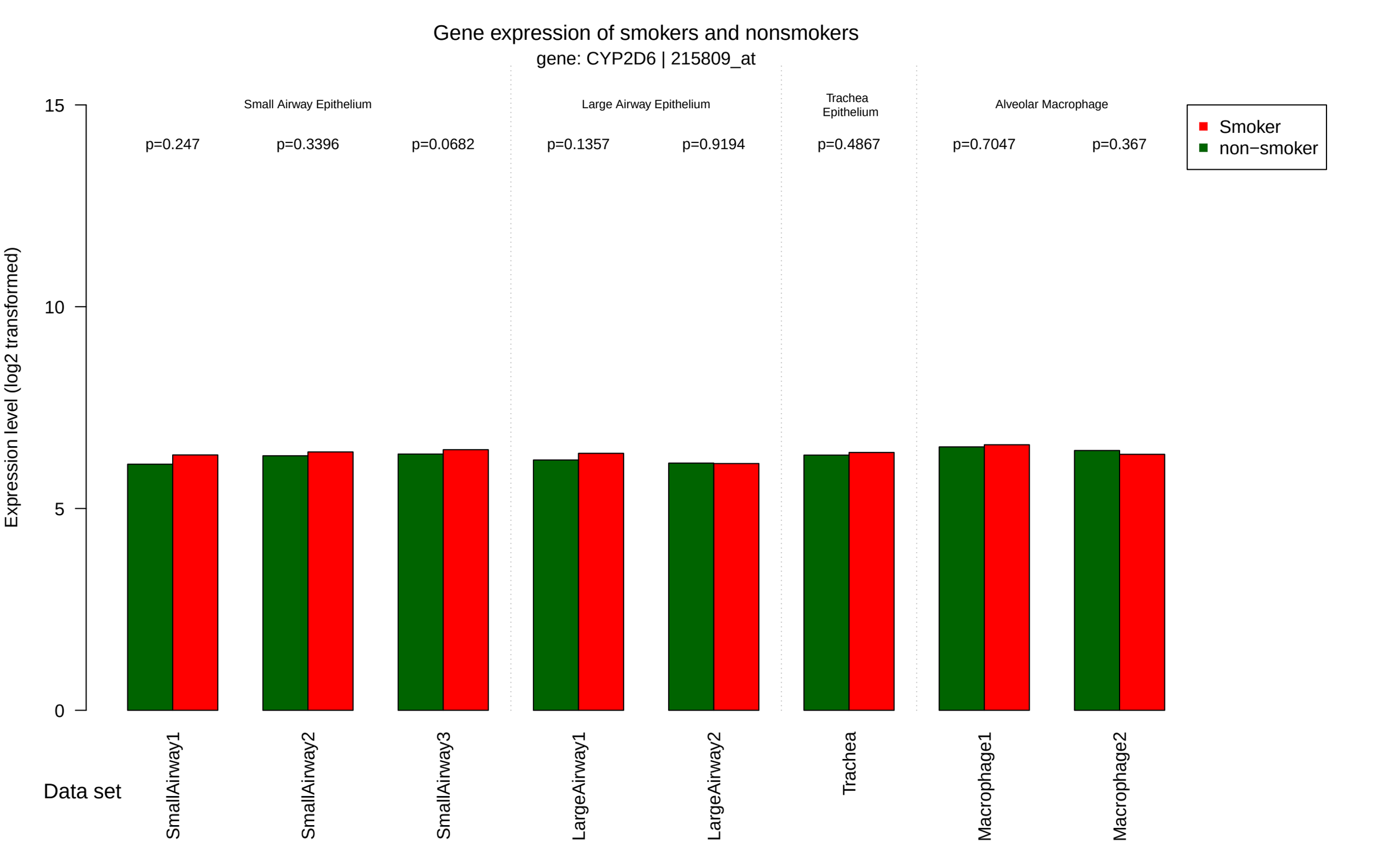
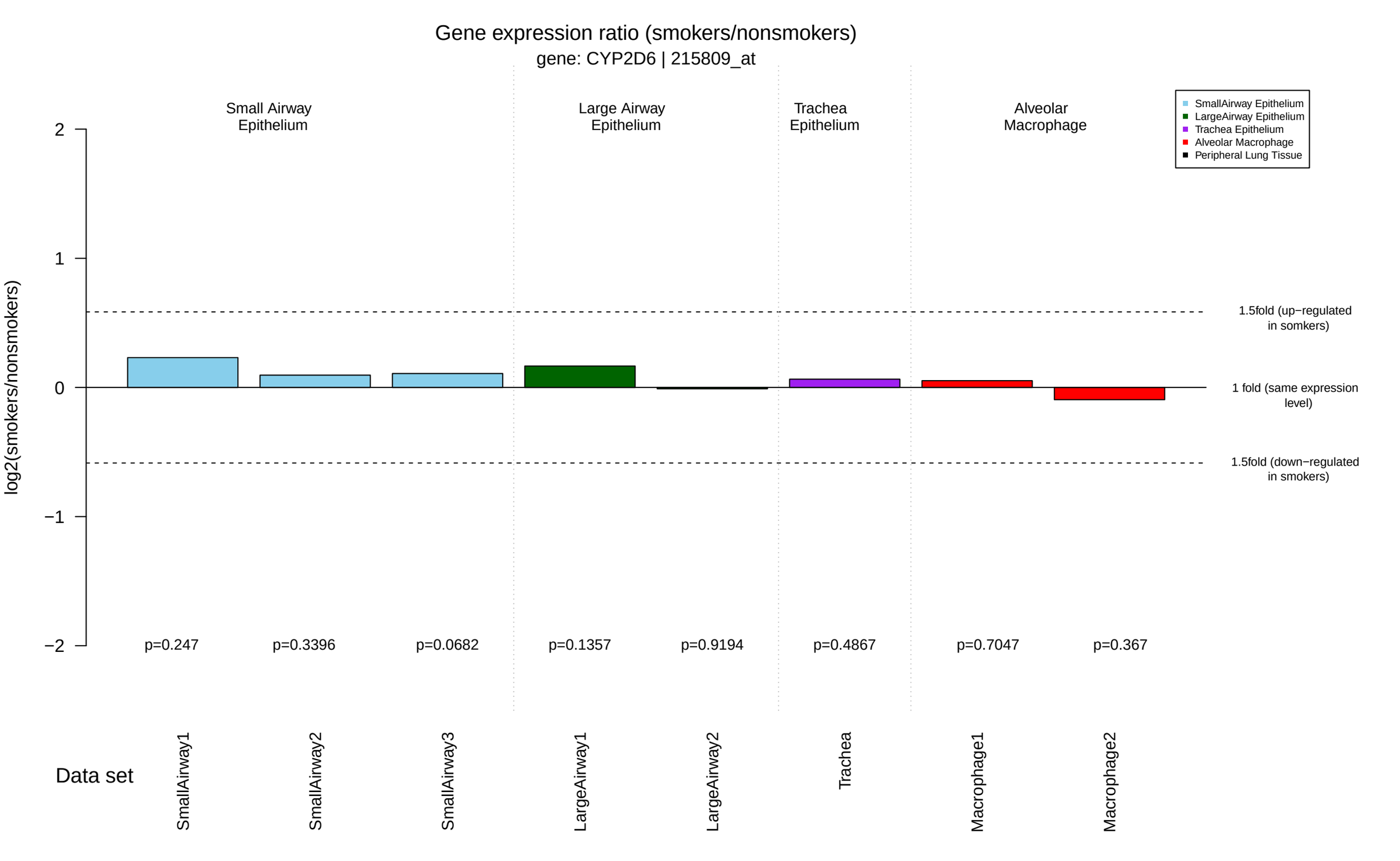
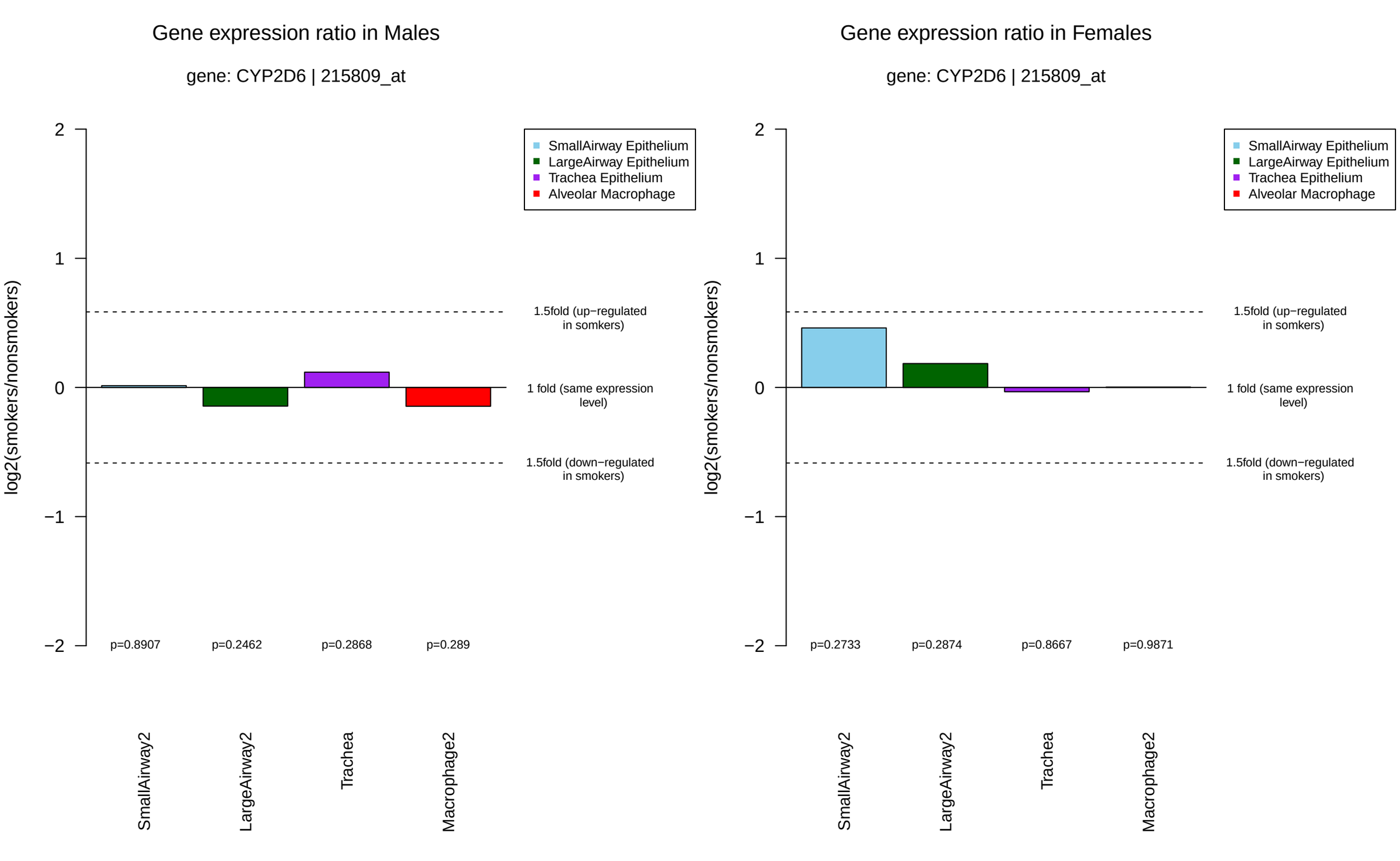
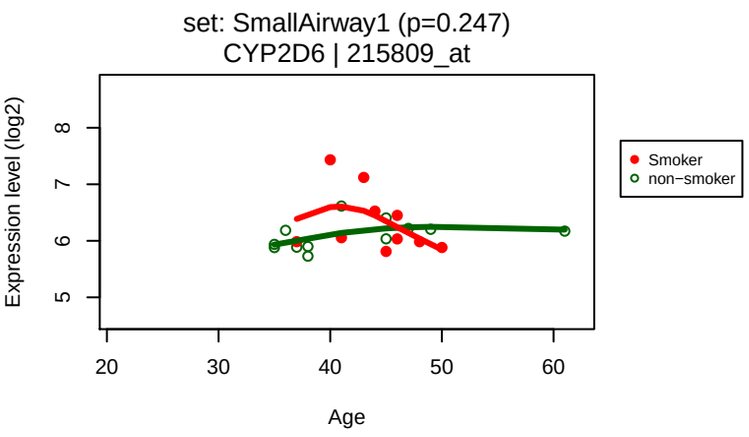
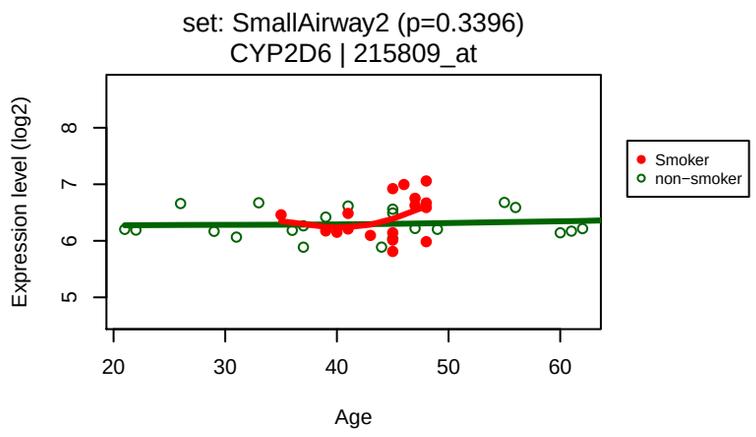
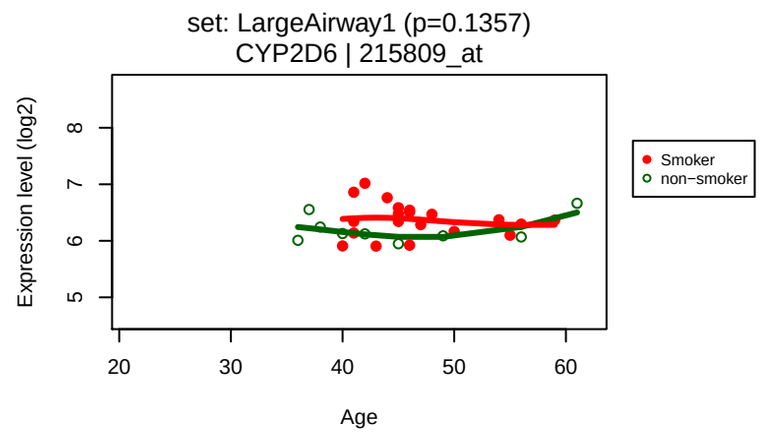

Gene expression of smokers and nonsmokers
gene: CYP2D6 | 215809_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.247
p=0.3396
p=0.0682
p=0.1357
p=0.9194
p=0.4867
p=0.7047
p=0.367
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

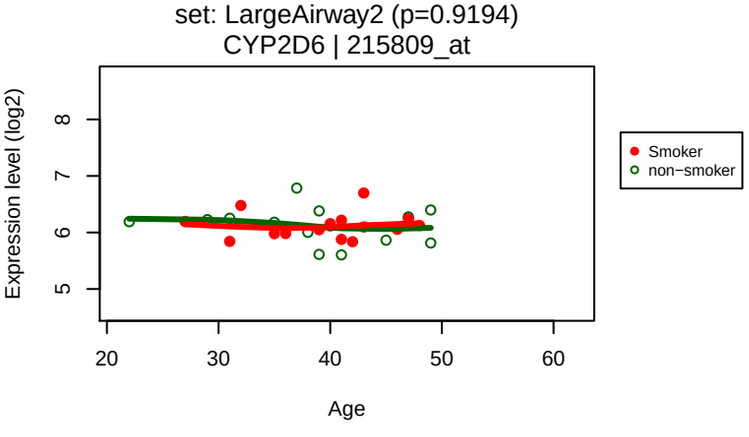

Gene expression ratio (smokers/nonsmokers)
gene: CYP2D6 | 215809_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.247
p=0.3396
p=0.0682
p=0.1357
p=0.9194
p=0.4867
p=0.7047
p=0.367
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

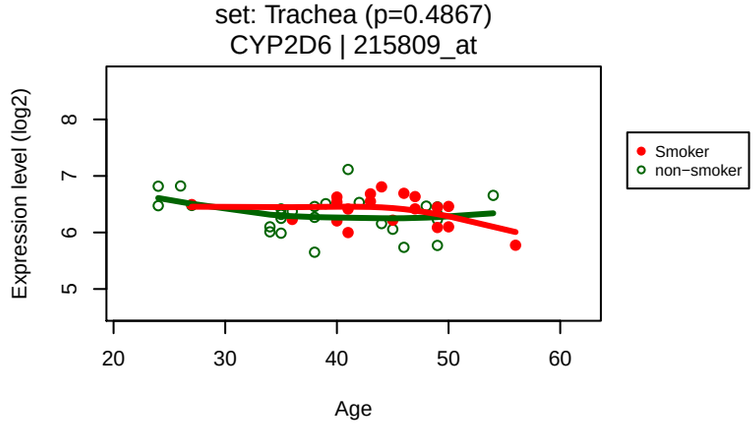

Gene expression ratio in Males
gene: CYP2D6 | 215809_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8907
p=0.2462
p=0.2868
p=0.289
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2D6 | 215809_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2733
p=0.2874
p=0.8667
p=0.9871
SmallAirway2
LargeAirway2
Trachea
Macrophage2

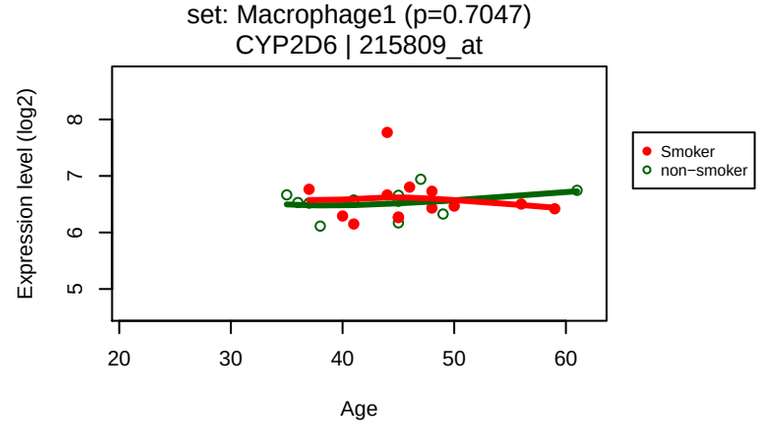

set: SmallAirway1 (p=0.247)
CYP2D6 | 215809_at
Expression level (log2)
Age
Smoker
non−smoker

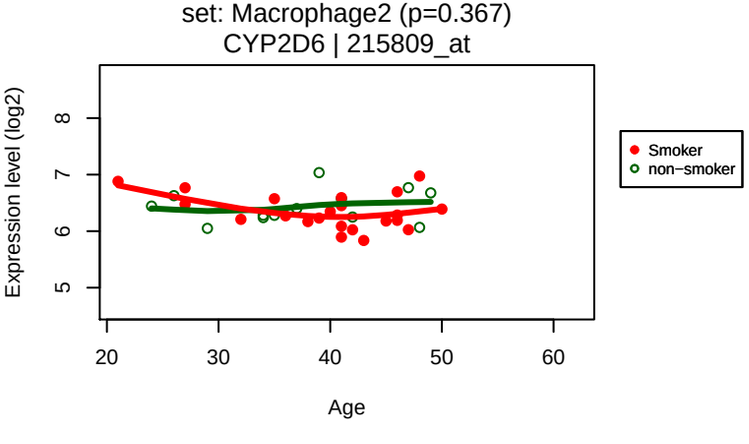

set: SmallAirway2 (p=0.3396)
CYP2D6 | 215809_at
Expression level (log2)
Age
Smoker
non−smoker

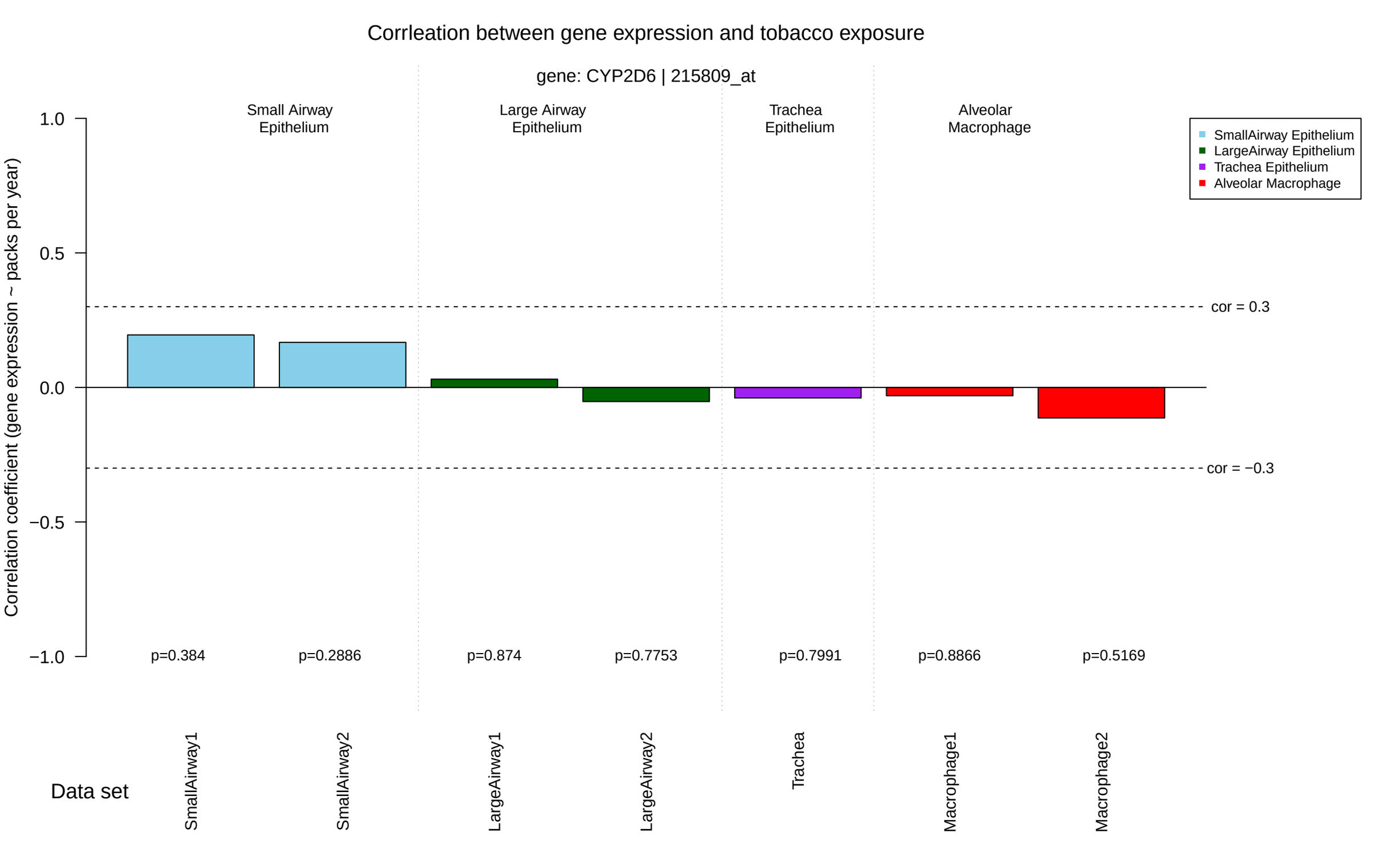

set: LargeAirway1 (p=0.1357)
CYP2D6 | 215809_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9194)
CYP2D6 | 215809_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4867)
CYP2D6 | 215809_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7047)
CYP2D6 | 215809_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.367)
CYP2D6 | 215809_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2D6 | 215809_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.384
p=0.2886
p=0.874
p=0.7753
p=0.7991
p=0.8866
p=0.5169
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2